Managing Sustainable Tourism in Juneau

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1 Abstract

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2 Introduction

2.1 Problem background

Juneau, Alaska, USA, is a popular tourist destination with over 30,000 residents and a record of more than 1.6 million cruise passengers. Tourism brings substantial revenue to the local community, while overcrowding, pressure on rainforest ecosystems, and concerns about glacier retreat highlight the need for sustainable tourism management.

Beyond its economic benefits, tourism carries **hidden costs** that affect infrastructure, the environment, and residents' quality of life. The following table summarizes these pressures alongside government responses to tourism-related challenges.

Tourism-related problems	**Government responses**
1. Pressure on **local infrastructure**, including water supply and waste management. 2. A larger **carbon footprint** and greater pressure on environmentally sensitive areas. 3. Reduced **quality of life for residents**, including housing shortages, overcrowding, and noise.	1. Increase **hotel taxes and visitor fees**. 2. **Cap** the number of daily visitors. 3. Restrict alcohol sales. 4. Use **additional revenue** to support conservation, improve infrastructure, and develop community assistance programs.

These measures motivate a modeling framework that considers both tourism benefits and the allocation of resources to mitigate its costs.

2.2 Problem statement

The central problem is to balance the economic benefits of tourism with environmental protection and residents' well-being. To support sustainable tourism management in Juneau, we address the following tasks:

1. **Develop a sustainable tourism model.** Define the objectives and constraints of tourism management, accounting for visitor numbers, tourism revenue, infrastructure capacity, and environmental protection. Use the model to propose management strategies and allocate tourism-related revenue.

2. **Identify key drivers through sensitivity analysis.** Evaluate how changes in model parameters affect the results and identify the factors that most strongly influence sustainable development, providing a basis for policy adjustments.

3. **Assess transferability to other destinations.** Adapt the model to destinations with different characteristics, evaluate its applicability, and explore strategies to promote less-visited attractions and achieve a more balanced distribution of tourism.

2.3 Our work

We will build an integrated modeling framework linking visitor volume, tourism revenue, environmental pressure, and community well-being. Within this framework, we will evaluate visitor limits, tourism-related taxes and fees, and the allocation of revenue to conservation, infrastructure, and community programs.

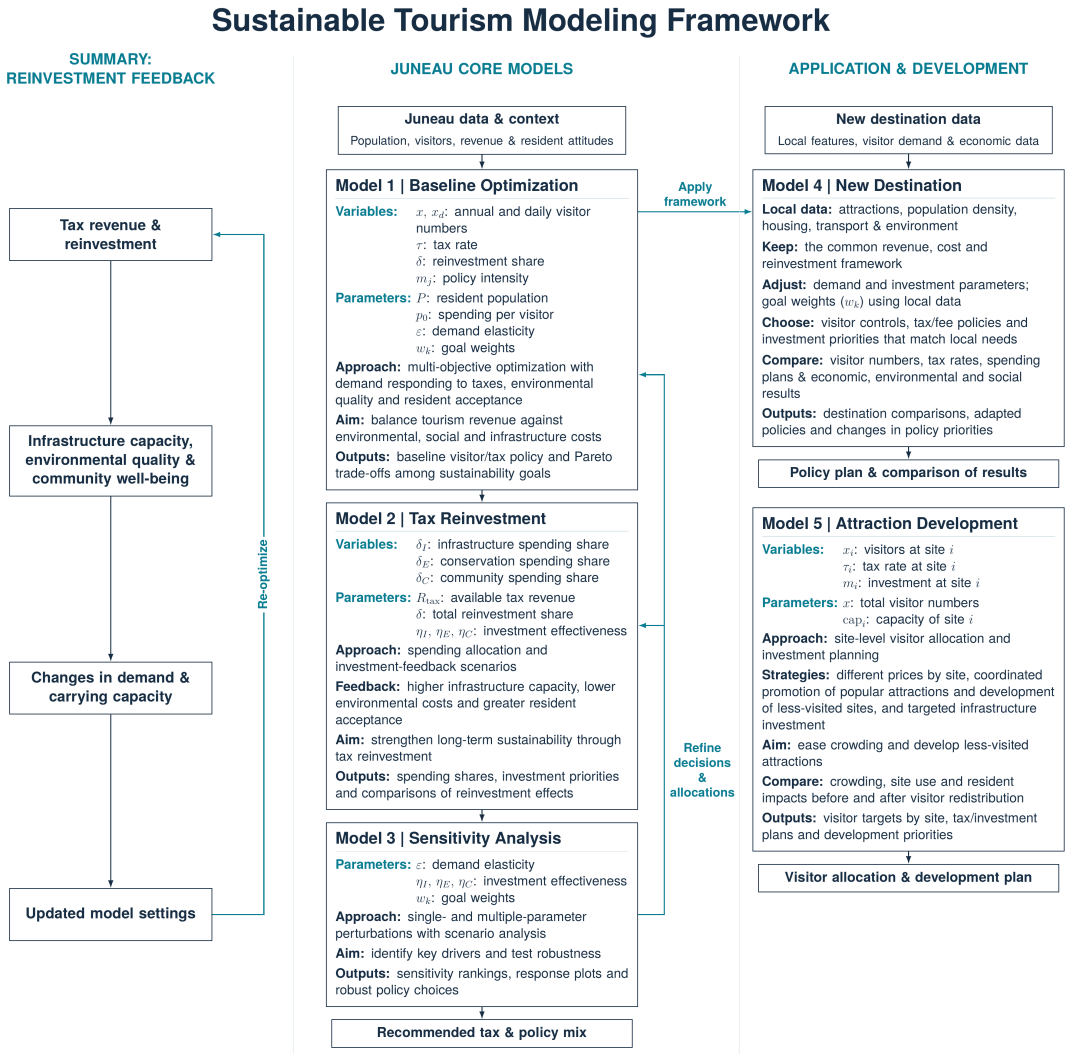


Figure 1: Work Framework

3 Assumptions and Justifications

Assumptions	Justifications
A1. *We regard some static variable as constant, and do not add any statistic which is not officially published*	*Holding baseline parameters fixed within each planning period allows us to isolate the effects of tourism policies. Their uncertainty can be examined through sensitivity analysis.*
A2. *We do not focus on extreme situation or accident incident*	*This assumption defines a normal-operation baseline for evaluating routine tourism policies. Major disruptions may alter demand and capacity substantially, requiring a separate scenario analysis.*
A3. *The government actions are reliable and immediately, means investment must have respond to model*	*The research does not design for predicting inability for government or direct agenda, but act as a general schedule.*

Assumptions	**Justifications**
A4. *Policies must meet ecological and visitor capacity limits, the five-ship daily cap, the budget limit, and a minimum annual tourism revenue of USD 360 million. Resident acceptance is a soft target.*	*These limits define a feasible policy. Any policy violating a hard limit is rejected. Resident acceptance below 0.7 is allowed, but lowers the policy score through a penalty.*
A5. *We model annual visitor volume x and peak daily visitor volume x_d separately, with a strict limit on daily visitors. Seasonal factors adjust capacity, without simulating each day.*	*This captures both yearly tourism activity and crowding on the busiest day. Detailed daily data are limited, and simulating every day would make the model harder to solve.*

4 Notations

The main symbols used throughout the modeling framework are listed below. Other symbols are defined where they first appear in each model.

Sets and Indices

Symbol	Definition
$j \in \mathcal{J}$	Management measure index
$k \in \mathcal{K}$	Investment project index
$i \in \mathcal{I}$	Tourism area index
$s \in \mathcal{S}$	Scenario index
$t \in \mathcal{T}$	Season or month index

Variables

Symbol	Meaning	Unit
x	Annual visitor volume	10^4 visitors/year
x_d	Peak daily visitor volume	10^4 visitors/day
I_{cap}	Annual infrastructure capacity	10^4 visitors/year
R	Tourism revenue	10^8 USD
B	Tax receipts available for allocation	10^8 USD
τ	Aggregate tourism tax rate	Dimensionless
$\delta_I, \delta_E, \delta_C$	Tax-revenue shares for infrastructure, environmental protection, and community programs	Dimensionless
$C_{env}, C_{soc}, C_{inf}$	Environmental, social, and infrastructure pressure indices, respectively	Dimensionless
S	Resident acceptance index	Dimensionless
J	Aggregate sustainability score	Dimensionless
m_j	Equals 1 if management measure j is adopted, and 0 otherwise	Dimensionless

Constants

Symbol	Value	Unit	Symbol	Value	Unit
P	3	10^4 residents	x_0	160	10^4 visitors
R_0	3.75	10^8 USD	p_0	0.0234375	10^8 USD per 10^4 visitors
x_{d0}	2	10^4 visitors/day	B_{phys}	7	ships/day
B_{pol}	5	ships/day	τ_0	0.11	Dimensionless

5 Models design

5.1 Data resources

We use the following baseline quantities for Juneau: 30,000 residents, 1.6 million cruise passengers in 2023, tourism revenue of USD 375 million, and a peak of about 20,000 visitors per day. The port's physical capacity of seven ships per day is distinguished from the policy limit of five. Visitor volumes are expressed in units of 10^4 people and monetary quantities in units of 10^8 USD. Thus, $p_0 = R_0/x_0 = 0.0234375$ in the model's units, equivalent to USD 234.375 per visitor. Demand elasticities, investment efficiencies, and policy-effect coefficients are modeling assumptions that require calibration and sensitivity analysis. All baseline statistics are provided in 2025 MCM Problem B.

5.2 Model 1: Baseline optimization

We use a mixed-integer nonlinear model to balance tourism revenue, environmental pressure, social costs, and infrastructure use. Tax receipts support reinvestment, which changes capacity and resident acceptance and feeds back into visitor demand.

5.2.1 Decision variables and model structure

Continuous variables describe visitor volume, tax rates, and spending shares. Integer and binary variables describe ship arrivals and management measures.

Symbol	Role	Meaning
x	Continuous	Annual visitor volume, in 10^4 visitors/year, subject to demand equilibrium
x_d	Continuous	Peak daily visitor volume, in 10^4 visitors/day
τ	Continuous	Aggregate tourism tax rate
$\delta_I, \delta_E, \delta_C$	Continuous	Tax-revenue shares for infrastructure, environment, and community programs
n_s	Integer	Number of ships on the modeled peak day
m_1	Binary	Adopt a stricter daily visitor cap
m_2	Binary	Restrict alcohol sales
m_3	Binary	Provide shore power
m_4	Binary	Manage traffic
m_5	Binary	Use marketing to redistribute visitors

Section 5.2.2 links revenue and demand. Section 5.2.3 calculates investment effects and pressure indices. Section 5.2.4 checks hard constraints, and Section 5.2.5 ranks feasible policies. The demand equation and state updates are solved together.

5.2.2 Revenue and demand response

Tourism revenue and tax receipts.

Symbol	Meaning
R, B	Tourism revenue and tax receipts, in 10^8 USD
$p_0 = 0.0234375$	Fixed spending per visitor in the model's units
$B_0 = 0.4125$	Reference tax receipts, calculated as $\tau_0 p_0 x_0$

$$R = p_0 x, \qquad B = \tau p_0 x. \tag{M1-1}$$

The term $p_0 x$ gives total tourism revenue by multiplying visitor volume by spending per visitor. The term $\tau p_0 x$ gives the government's tax receipts. The tax rate τ also affects x through demand, so revenue and tax receipts change with visitor numbers.

Demand response.

Symbol	Meaning
$x_0 = 160,\ \tau_0 = 0.11$	Reference annual visitor volume and tax rate
I_{cap}, I_0	Available and reference annual infrastructure capacity
$S, S_0 = 1$	Resident acceptance score and its reference level
$\varepsilon = -0.5,\ \alpha = \beta = 0.2$	Tax, capacity, and acceptance elasticities
(κ_1, κ_2)	Demand effects of restrictions: $(-0.1, -0.05)$
$(\kappa_3, \kappa_4, \kappa_5)$	Demand effects of improvements: $(0.02, 0.03, 0.1)$

$$x = x_0 \left(\frac{\tau}{\tau_0}\right)^{\varepsilon} \left(\frac{I_{cap}}{I_0}\right)^{\alpha} \left(\frac{\max\{0.01, S\}}{S_0}\right)^{\beta} \prod_{j=1}^{5} (1 + \kappa_j m_j). \tag{M1-2}$$

Here x_0 is the reference visitor volume. The tax factor $(\tau/\tau_0)^{\varepsilon}$ describes how higher taxes reduce demand. The capacity factor $(I_{cap}/I_0)^{\alpha}$ and acceptance factor $(\max\{0.01, S\}/S_0)^{\beta}$ describe how better facilities and resident acceptance attract visitors. The product $\prod_{j=1}^{5}(1 + \kappa_j m_j)$ combines the demand effects of adopted measures. These factors update the visitor volume used to calculate tax receipts and investment effects.

5.2.3 Investment effects and pressure indices

Infrastructure capacity.

Symbol	Meaning
$I_0 \approx 132.2314$	Reference annual capacity, in 10^4 visitors/year, calibrated using the earlier solver
$\eta_I = 0.7, \delta_I$	Infrastructure investment efficiency and spending share
B/B_0	Tax receipts relative to the reference level
$\delta_{I,0} = 0.3$	Reference infrastructure spending share used to calibrate I_0

$$I_{cap} = I_0 \left(1 + \eta_I \delta_I \frac{B}{B_0}\right). \tag{M1-3}$$

Here I_0 is the capacity before investment. The term $\eta_I \delta_I B/B_0$ is the proportional improvement from infrastructure spending, so $I_0 \eta_I \delta_I B/B_0$ is the added capacity. Following the earlier calibration, $I_0 = x_0/(1 + \eta_I \delta_{I,0}) = 160/1.21 \approx 132.2314$.

Environmental pressure.

Symbol	Meaning
C_{env}	Environmental-pressure index
$x_0 = 160,\ x_{d0} = 2$	Reference annual and peak daily visitor volumes
$\eta_E = 0.7, \delta_E$	Environmental investment efficiency and spending share
0.8, 0.4	Reference environmental pressure and assumed residual-pressure floor
$0.3m_3$	A 30% reduction in the peak-day pressure component when shore power is adopted

$$C_{env} = \max\left\{0.4,\ 0.8\left[0.5\frac{x}{x_0} + 0.5\frac{x_d}{x_{d0}}(1 - 0.3m_3)\right] - \eta_E\delta_E\frac{B}{B_0}\right\}. \tag{M1-4}$$

The term $0.8(0.5x/x_0)$ represents pressure from annual visitor activity. The term $0.8(0.5x_d/x_{d0})(1 - 0.3m_3)$ represents peak-day pressure, with $1 - 0.3m_3$ capturing the reduction from shore power. The term $-\eta_E\delta_E B/B_0$ represents the benefit of environmental investment. The maximum operator accounts for environmental pressure that remains after mitigation.

Social costs.

Symbol	Meaning
C_{soc}	Scaled social-cost index
$P = 3$	Resident population, in 10^4 residents
$C_{soc,0} = 2844$	Reference divisor specified in the source model
0.5, 0.3, 0.2	Weights for population pressure, congestion, and noise
$0.4m_4,\ 0.2m_2$	Congestion and noise reduction fractions under the corresponding measures

$$C_{soc} = \frac{1}{C_{soc,0}}\left[0.5\left(\frac{x}{P}\right)^2 + 0.3\frac{x_d}{x_{d0}}(1 - 0.4m_4) + 0.2(1 - 0.2m_2)\right]. \tag{M1-5}$$

The term $0.5(x/P)^2$ represents crowding costs that grow faster as visitor numbers rise. The term $0.3(x_d/x_{d0})(1 - 0.4m_4)$ represents congestion, reduced by traffic management. The term $0.2(1 - 0.2m_2)$ represents noise, reduced by alcohol restrictions. Multiplying their sum by $1/C_{soc,0}$ rescales social costs for the objective.

Infrastructure pressure.

Symbol	Meaning
C_{inf}	Scaled infrastructure-pressure index
$C_{inf,0} = 1137$	Reference divisor specified in the source model
0.4, 0.2	Weights for annual capacity use and peak-day arrivals

$$C_{inf} = \frac{1}{C_{inf,0}}\left[0.4\left(\frac{x}{I_{cap}}\right)^2 + 0.2\frac{x_d}{x_{d0}}\right]. \tag{M1-6}$$

The term $0.4(x/I_{cap})^2$ represents pressure from annual infrastructure use, which grows faster as visitor numbers rise. The term $0.2x_d/x_{d0}$ represents pressure from peak-day arrivals. Multiplying their sum by $1/C_{inf,0}$ rescales infrastructure costs for the objective.

Resident acceptance.

Symbol	Meaning
S	Resident acceptance score relative to a reference level of 1.0
$x/x_0 - 1,\ x_d/x_{d0} - 1$	Changes in annual activity and daily crowding relative to the reference
$\eta_C = 0.7, \delta_C$	Community investment efficiency and spending share
$0.4m_4,\ 0.1m_5$	Traffic-management reduction fraction and redistribution benefit

$$S = 1.0 - 0.3\left(\frac{x}{x_0} - 1\right) - 0.1\left(\frac{x_d}{x_{d0}} - 1\right)(1 - 0.4m_4) + \eta_C\delta_C\frac{B}{B_0} + 0.1m_5. \tag{M1-7}$$

The constant 1.0 is the reference acceptance score. The term $-0.3(x/x_0 - 1)$ describes the effect of changes in annual visitor volume: more visitors lower acceptance, while fewer visitors raise it. The term $-0.1(x_d/x_{d0} - 1)(1 - 0.4m_4)$ describes the effect of changes in daily crowding, with $1 - 0.4m_4$ reducing this adjustment under traffic management. The term $\eta_C\delta_C B/B_0$ adds the benefit of community investment, and $0.1m_5$ adds the benefit of visitor redistribution.

5.2.4 Hard constraints

M1 enforces visitor capacity, environmental protection, tax bounds, a minimum revenue, the spending budget, and daily ship limits.

Visitor capacity.

Symbol	Meaning
I_{cap}	Available annual infrastructure capacity
$x_{d0} = 2$	Reference daily visitor limit, in 10^4 visitors/day
$\Delta_{cap} = 0.8$	Reduction in the daily cap if $m_1 = 1$, in 10^4 visitors/day

$$0 \le x \le I_{cap}, \qquad 0 \le x_d \le x_{d0} - \Delta_{cap}m_1. \tag{M1-8}$$

The condition $x \le I_{cap}$ links annual visitor volume to available infrastructure. In the daily-cap expression $x_{d0} - \Delta_{cap}m_1$, the term $\Delta_{cap}m_1$ represents the reduction caused by adopting the stricter visitor-limit measure.

Environmental, tax, and revenue limits.

Symbol	Meaning
0.95	Maximum environmental-pressure index
0.09, 0.15	Minimum and maximum tax rates
3.6	Minimum tourism revenue, in 10^8 USD

$$C_{env} \le 0.95, \qquad 0.09 \le \tau \le 0.15, \qquad R \ge 3.6. \tag{M1-9}$$

The conditions on C_{env}, τ, and R enforce ecological protection, an admissible tax policy, and the required tourism income. All three are hard requirements when screening policies.

Investment budget.

Symbol	Meaning
$\delta_I, \delta_E, \delta_C$	Nonnegative shares of the same tax receipts
0.8	Maximum total share available for these investments

$$\delta_I + \delta_E + \delta_C \le 0.8, \qquad \delta_I, \delta_E, \delta_C \ge 0. \tag{M1-10}$$

The sum $\delta_I + \delta_E + \delta_C$ combines the shares allocated to infrastructure, environmental protection, and community programs. The budget constraint ensures that these three areas draw from the same available tax receipts.

Ship arrivals.

Symbol	Meaning
n_s	Nonnegative integer number of daily ship arrivals
$\bar{c} = 0.4$	Assumed average passengers per ship, in 10^4 visitors/ship, derived as $x_{d0}/5$
5	Policy maximum number of ships per day

$$x_d \le \bar{c}\, n_s, \qquad n_s \le 5, \qquad n_s \in \{0, 1, \ldots, 5\}. \tag{M1-11}$$

The product $\bar{c}\, n_s$ converts the ship count into daily passenger capacity. The condition $x_d \le \bar{c}\, n_s$ connects peak visitor arrivals to that capacity, while the condition on n_s applies the daily ship policy.

Management measures.

Symbol	Meaning
m_j	Adoption indicator for management measure j, defined in Section 5.2.1

$$m_j \in \{0, 1\}, \qquad j = 1, \ldots, 5. \tag{M1-12}$$

Each m_j switches a management measure on or off. Its associated terms, such as $\kappa_j m_j$ in demand and $\Delta_{cap} m_1$ in the daily cap, apply the effect of that choice.

5.2.5 Sustainability score

Symbol	Meaning
J	Dimensionless sustainability score
R/R_0	Revenue relative to the reference $R_0 = 3.75$
$(0.4, 0.3, 0.2, 0.1)$	Assumed weights for revenue, environment, society, and infrastructure
0.7, 5.0	Acceptance target and penalty coefficient

$$\begin{aligned} \max \; J = 0.4\frac{R}{R_0} - 0.3C_{env} - 0.2C_{soc} - 0.1C_{inf} \\ - 5.0 \max\{0, 0.7 - S\}. \end{aligned} \tag{M1-13}$$

The term $0.4R/R_0$ rewards tourism revenue relative to the reference. The terms $-0.3C_{env}$, $-0.2C_{soc}$, and $-0.1C_{inf}$ subtract environmental, social, and infrastructure costs. The term $-5.0 \max\{0, 0.7 - S\}$ subtracts a penalty for unmet resident acceptance. Together, these terms rank feasible policies by their economic benefits and tourism pressures.

5.2.6 Solution procedure and calibration checks

1. **Enumerate discrete choices:** consider the 32 measure combinations and admissible integer ship counts.
2. **Solve the demand feedback:** for each candidate tax rate, peak volume, and spending allocation, iterate revenue, capacity, acceptance, and demand until visitor volume is consistent. Then recalculate all pressure indices using the final volume.
3. **Check feasibility and compare:** enforce M1-8–M1-12 and rank feasible candidates by M1-13. Infeasible cases must be reported separately. A local nonlinear search does not establish global optimality.